

CAPSTONE PROJECT

WEBSITE – SAUCE DEMO

Link: <https://www.saucedemo.com>

About

Sauce Demo is a sample e-commerce web application used for practicing and demonstrating test automation skills. It simulates a real online shopping experience, including login, product browsing, cart management, and checkout flow. It is widely used for automation testing practice because of its stable and predictable behavior.

Objective

This project aims to automate and test the Sauce Demo website using the Playwright automation framework.

The main goal is to validate core e-commerce workflows such as login, product selection, cart operations, checkout process, and user session handling.

Automation helps reduce manual effort, improve testing speed, increase accuracy, and ensure application reliability.

Tech Stack

Category	Technology
Automation Framework	Playwright
Programming Language	JavaScript
Runtime Environment	Node.js
IDE	VS Code
Package Manager	npm
Browser Support	Chromium, Firefox, WebKit
Reporting Tool	Allure Report
CI/CD Integration	GitHub Actions
Containerization	Docker

Core Functionalities

1. User login and session handling
2. Product listing and product details
3. Product sorting and filtering
4. Add to cart and remove from cart
5. Cart quantity update and validation
6. Checkout workflow
7. Order confirmation process
8. User session logout

Services

1. Authentication Service
2. Product Service
3. Search Service
4. Profile Service
5. Cart Management Service
6. Wishlist Service
7. Checkout / Order Processing Service
8. Order Summary Service

Scope of Testing

The scope of this capstone project is limited to the frontend functionality of the Sauce Demo website. The focus is on functional UI automation using Playwright.

It excludes backend validation, database testing, and performance or security testing.

In Scope

- User login and logout functionality
- Product listing and product details validation
- Add to cart, remove from cart, and cart updates

- Checkout flow and order placement simulation
- Session handling and navigation flow

Out of Scope

- Real payment processing
- Backend/database validation
- Performance, load, and stress testing
- Security testing
- Server-side configuration testing

Service-Wise Test Case Distribution

Each service will contain 15 functional test cases, resulting in a total of 120 test cases.

No	Service / Module	Testing Coverage	Count
1	Authentication Service	Login, logout, invalid credentials, session validation, locked user handling, empty fields, password checks	15
2	Product Service	Product listing, product details, sorting, filtering, price validation, image validation, navigation	15
3	Search Service	Product search, empty search, invalid search, result validation, performance behavior, reset search	15
4	Profile Service	User info handling, session persistence, navigation validation, logout state handling, user data display	15
5	Cart Management Service	Add/remove items, quantity update, cart badge validation, subtotal check, persistence, multiple items handling	15
6	Wishlist Service	Add/remove items, duplicate handling, wishlist persistence, move to cart, UI validation	15
7	Checkout / Order Processing Service	Checkout flow, order summary, form validation, step navigation, error handling, confirmation page	15
8	Order Summary Service	Order confirmation, final review, price validation, success message, navigation after order	15

Test Data Strategy

Since Sauce Demo is a public test application:

- Use standard test users provided by the system
 - Avoid real personal or financial data
 - Use reusable dummy data for automation flows
 - Maintain separate test data files for scalability
 - Ensure consistent test execution across environments
 - Reset state between test runs when required
-

Project Goals

- Build a scalable Playwright automation framework
 - Cover 120 functional test cases across 8 services
 - Validate end-to-end e-commerce workflows
 - Integrate Allure reporting for test visualization
 - Implement CI/CD using GitHub Actions
 - Enable Docker-based execution support
 - Follow industry-level automation practices
 - Improve maintainability and reusability of test scripts
-

Conclusion

This capstone project focuses on automating the Sauce Demo website using Playwright to validate real-world e-commerce workflows such as authentication, product browsing, cart management, and checkout processing.

The project emphasizes functional testing with a structured framework that includes CI/CD integration, Allure reporting, and Docker support. It aims to demonstrate strong automation skills by building a scalable and maintainable testing architecture suitable for real-world QA environments.